AISI 1010 Steel, cold drawn

Categories:	Metal; Ferrous Metal; Carbon Steel; AISI 1000 Series Steel; Low Carbon Steel
Material Notes:	Used widely in low strength applications. Good formability, fair machinability, can be hardened by cyaniding. Suitable for magnet core applications
Key Words:	UNS G10100, AISI1010
Vendors:	No vendors are listed for this material. Please click here if you are a supplier and would like information on how to add your listing to this material.

Physical Properties	**Metric**	**English**	**Comments**
Density	7.87 g/cc	0.284 lb/in³	

Mechanical Properties	**Metric**	**English**	**Comments**
Hardness, Brinell	105	105	
Hardness, Knoop	123	123	Converted from Brinell
Hardness, Rockwell B	60	60	Converted from Brinell
Hardness, Vickers	108	108	Converted from Brinell
Tensile Strength, Ultimate	365 MPa	52900 psi	
Tensile Strength, Yield	305 MPa	44200 psi	
Elongation at Break	20 %	20 %	In 50 mm
Reduction of Area	40 %	40 %	
Modulus of Elasticity	205 GPa	29700 ksi	Typical for steel
Bulk Modulus	160 GPa	23200 ksi	Typical for steel
Poissons Ratio	0.29	0.29	Typical For Steel
Machinability	55 %	55 %	Based on AISI 1212 steel = 100%. Group I bar, rod, and wire products machinability can be improved by cold drawing.
Shear Modulus	80.0 GPa	11600 ksi	Typical for steel

Electrical Properties	**Metric**	**English**	**Comments**
Electrical Resistivity	0.0000143 ohm-cm	0.0000143 ohm-cm	condition unknown

Thermal Properties	**Metric**	**English**	**Comments**
CTE, linear	12.2 µm/m-°C @Temperature 0.000 - 100 °C	6.78 µin/in-°F @Temperature 32.0 - 212 °F	
	13.5 µm/m-°C @Temperature 0.000 - 300 °C	7.50 µin/in-°F @Temperature 32.0 - 572 °F	
	14.2 µm/m-°C @Temperature 0.000 - 500 °C	7.89 µin/in-°F @Temperature 32.0 - 932 °F	
Specific Heat Capacity	0.448 J/g-°C @Temperature >=100 °C	0.107 BTU/lb-°F @Temperature >=212 °F	condition unknown
	0.498 J/g-°C @Temperature 150 - 200 °C	0.119 BTU/lb-°F @Temperature 302 - 392 °F	
	0.519 J/g-°C @Temperature 200 - 250 °C	0.124 BTU/lb-°F @Temperature 392 - 482 °F	

	0.536 J/g-°C @Temperature 250 - 300 °C	0.128 BTU/lb-°F @Temperature 482 - 572 °F	
	0.565 J/g-°C @Temperature 300 - 350 °C	0.135 BTU/lb-°F @Temperature 572 - 662 °F	
	0.590 J/g-°C @Temperature 350 - 400 °C	0.141 BTU/lb-°F @Temperature 662 - 752 °F	
	0.649 J/g-°C @Temperature 400 - 450 °C	0.155 BTU/lb-°F @Temperature 752 - 842 °F	
	0.729 J/g-°C @Temperature 550 - 600 °C	0.174 BTU/lb-°F @Temperature 1020 - 1110 °F	
	0.825 J/g-°C @Temperature 650 - 700 °C	0.197 BTU/lb-°F @Temperature 1200 - 1290 °F	
Thermal Conductivity	49.8 W/m-K	346 BTU-in/hr-ft²-°F	Typical steel

Component Elements Properties	Metric	English	Comments
Carbon, C	0.080 - 0.13 %	0.080 - 0.13 %	
Iron, Fe	99.18 - 99.62 %	99.18 - 99.62 %	As remainder
Manganese, Mn	0.30 - 0.60 %	0.30 - 0.60 %	
Phosphorus, P	<= 0.040 %	<= 0.040 %	
Sulfur, S	<= 0.050 %	<= 0.050 %	

References for this datasheet.